

TASK 17: #51617

1. create a script on your instance which writes the disk usage to a file and send an email to specific mail.
2. verify the recipient mail id in SNS console
3. configure simple notification service[SNS] For delivering the mail notifications.
4. verify the mail sent

Step 1 — Verify the Recipient Email in the SNS Console

Create the SNS topic

1. Open the SNS console and select Topics in the left sidebar.
2. Click Create topic.
3. Choose type: Standard.
4. Enter a name (e.g. diskspace).
5. Click Create topic, then copy the Topic ARN shown on the resulting page.

Before any email can be delivered, the recipient address must be subscribed to the SNS topic and must confirm that subscription.

6. Open the SNS console and go to Subscriptions.
7. Locate the subscription for the recipient email address and check the Status column.
8. If Status shows "Pending confirmation", open the recipient's inbox for an email from AWS Notifications with the subject "AWS Notification – Subscription Confirmation", and click Confirm subscription.
9. Refresh the Subscriptions table — the Status should now read "Confirmed".

Common issue: a mistyped email address (e.g. "gamil.com" instead of "gmail.com") will leave the subscription stuck in Pending confirmation indefinitely, since the confirmation email can never be delivered. Delete any mistyped subscriptions and recreate them with the correct address.

Step 2 — Configure Amazon SNS for Email Delivery

Subscribe an email address

10. On the topic's page, scroll to Subscriptions and click Create subscription.
11. Set Protocol to Email.
12. Enter the recipient's email address under Endpoint.
13. Click Create subscription, then confirm it as described in Step 2.

Grant the EC2 instance permission to publish (IAM console)

Without this step, the script will fail with a "NoCredentials" error when it tries to call SNS. Create and attach the role entirely through the console:

14. Open the IAM console and select Roles, then click Create role.
15. For Trusted entity type, choose AWS service; for Use case, choose EC2. Click Next.
16. Click Create policy (opens a new tab). Select the JSON tab and paste the policy below.

```
{  
  "Version": "2012-10-17",
```

```

"Statement": [
  {
    "Effect": "Allow",
    "Action": "sns:Publish",
    "Resource": "arn:aws:sns:ap-south-1:176938214725:diskspace"
  }
]
}

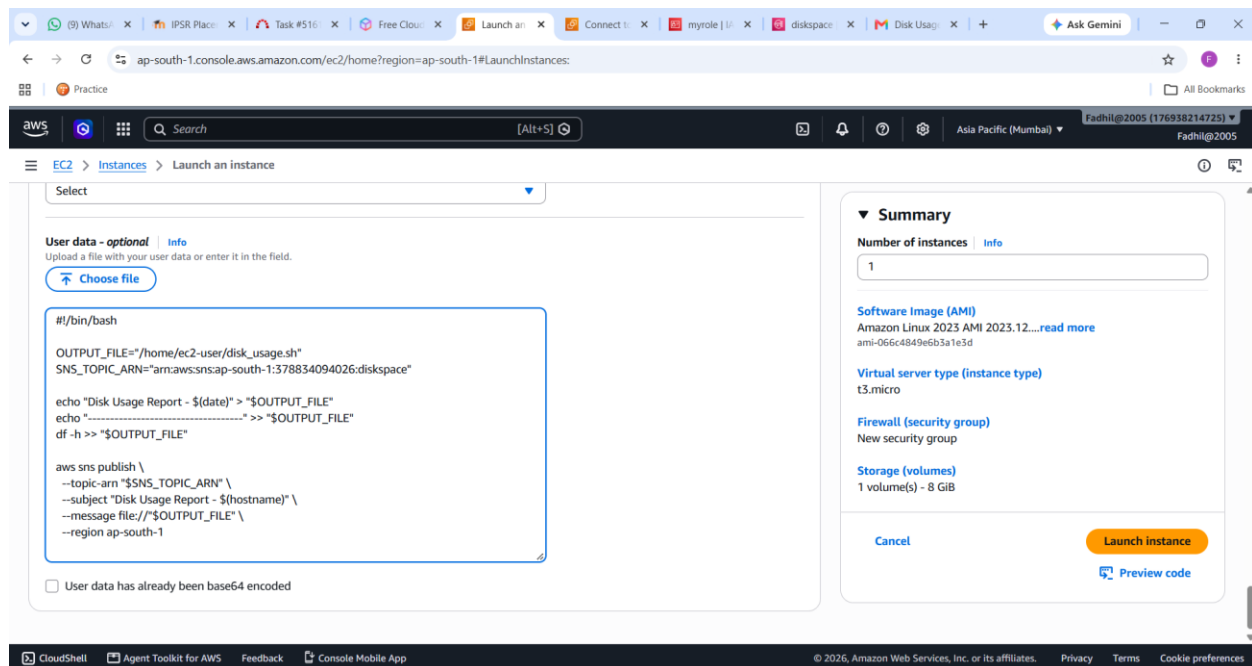
```

17. Click Next, name the policy (e.g. SNSPublishPolicy), and click Create policy.
 18. Return to the role-creation tab, refresh the policy list, select SNSPublishPolicy, and click Next.
 19. Name the role (e.g. DiskUsageEC2Role) and click Create role.
 20. Go to the EC2 console, select Instances, select your instance, then Actions → Security → Modify IAM role.
 21. Choose DiskUsageEC2Role from the dropdown and click Update IAM role.
- This can be done on a running instance without stopping or restarting it. Allow 10–15 seconds for the role to propagate before the instance can use it.

Step 3 — Create the Disk Usage Script (via EC2 Instance Connect)

22. Open the EC2 console and select Instances.
23. Select your instance, click Connect (top right).
24. Choose the EC2 Instance Connect tab and click Connect.

This opens a terminal tab directly in your browser, logged in as ec2-user — no SSH keys or external tools needed. Create the script in this terminal:



Step 4 — Run the Script and Verify the Email

Back in the same EC2 Instance Connect browser terminal from Step 1:

25. Run the script with verbose output to catch any errors:

```
bash -x ~/disk_usage.sh
```

OUTPUT:

